Mock Referee Report — JASA

Permutations accelerate Approximate Bayesian Computation

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Recommendation: Major revision.

Summary

The paper introduces PERMABC, an ABC framework for hierarchical models with exchangeable compartments. The core idea—solving a linear sum assignment problem to optimally match simulated compartments to observed ones before checking the ABC criterion—is elegant and practically relevant. The authors develop sequential extensions (PERMABC-SMC), as well as two novel bridging strategies (Over-Sampling and Under-Matching), and demonstrate the approach on a COVID-19 epidemic model with 94 French departments ($d = 283$).

The contribution is original and the practical motivation is strong. However, the paper suffers from a gap between its ambitions and what is actually demonstrated. The theoretical results are limited in scope, the numerical evaluation relies almost exclusively on a single proxy metric (tolerance ε), and several claims lack adequate empirical or theoretical support.

Major concerns

M1. The sole evaluation metric is ε—this is insufficient.

Throughout Sections 3.2, 3.4, and Appendix F, every comparison between methods is presented as "ε achieved vs. computational cost." But ε is a tuning parameter of the algorithm, not a measure of inferential quality. A method that reaches a smaller ε faster is not necessarily producing a better posterior approximation—it could be concentrating around a wrong mode, or suffering from particle degeneracy that is not reflected in ε.

A JASA reader will expect at least one proper measure of posterior quality: marginal KL divergence against a reference posterior (available in closed form for the Gaussian toy model), Wasserstein distance, coverage of credible intervals, or squared error on point estimates. Without this, the central claim—that PERMABC methods are "more efficient"—is only partially supported.

M2. No comparison with standard ABC-SMC (without permutations).

The paper compares PERMABC-SMC with Vanilla ABC, ABC-PMC, and ABC-Gibbs. But the natural baseline—standard ABC-SMC with the same adaptive ε schedule of Del Moral et al. (2012), just without the permutation step—is absent. This makes it impossible to isolate the contribution of the permutation idea from the choice of SMC framework. The first question any reviewer will ask is: "How much of the gain comes from permutations, and how much from switching to SMC?"

M3. Theorem 1 is correct but limited, and the paper oversells it.

Theorem 1 states that PERMABC = ABC when $\varepsilon < \varepsilon^*$. This is clean but says little about the practically relevant regime where $\varepsilon \geq \varepsilon^*$. The paper claims "the efficiency gains provided by PERMABC come at no cost to theoretical consistency" (Section 1.4), but this is only true asymptotically as $\varepsilon \to 0$. For any fixed ε used in practice, the two posteriors may differ, and there is no bound on the discrepancy. The Corollary is a direct consequence and adds no new content. I would suggest either:

- providing a finite-ε bound on the TV or Wasserstein distance between π^*_ε and π_ε, or
- toning down the claims and clearly stating the limitation.

M4. The Over-Sampling asymptotic results ($M \to \infty$) are stated informally.

The new content in Section 2.3 (pushforward formulation, $M \to \infty$ convergence) is the most theoretically interesting part of the paper. However, the results are presented as displayed equations rather than as formal propositions with explicit assumptions and proofs. The key convergence claim ($d_M \to \delta_\beta$ a.s.) relies on "mild regularity assumptions on d" that are never stated. For JASA, these results should be either:

- formalized as proper propositions with proofs (even short ones), or
- clearly labeled as heuristic/conjectural with a pointer to where rigorous treatment will appear.

I note that the authors appear to have additional material with a more complete treatment including a $K \to \infty$ WABC connection. This would substantially strengthen the paper and should be integrated.

M5. The SIR application is a proof of concept, not a validation.

The SIR section is well-written and honest about model misspecification. However:

- There is no comparison with any other method on this problem (not even standard ABC-SMC). The reader cannot assess whether permABC is actually better here, or just the only thing that was tried.
- The noise model (log-normal on δ with $\sigma = 0.05$) is introduced but not justified. Why this form? Why this value of σ? Was there a sensitivity analysis?
- The posterior on ν_k reaching values of 4 is acknowledged as implausible. This raises the question: is permABC producing a good approximation of a bad model, or a bad approximation? Without a simulated-data experiment where the ground truth is known, it is impossible to tell.
- The claim that "permABC was specifically designed for these challenges" positions the SIR as the motivating application, yet the numerical evidence is limited to two figures (posterior of R_0 and a map). At minimum, I would expect posterior predictive checks or a simulation study on synthetic SIR data with known parameters.

M6. No numerical comparison with Wasserstein ABC.

The revised manuscript now clarifies in Section 1.2 and in an asymptotic appendix that permABC is a hierarchical analogue of WABC at the compartment level, and coincides with it in the one-dimensional case. These connections are textual only: no experiment on the Gaussian toy model compares the two approaches in terms of accuracy, runtime, or robustness, despite the claim that exact LSA is preferable in the regime of moderate K and non-trivial per-compartment dimension. A direct benchmark against WABC's Hilbert-based and Sinkhorn-based approximations—on the toy Gaussian model where the posterior is closed-form—would substantiate this claim and address a point raised by earlier readers of this manuscript.

M7. Lemma 1 and Theorem 1 should be extended to the case of ties.

The current statement of Lemma 1 assumes that no pair of observed compartments have exactly equal data. In practice, even for continuous data, observations that are numerically close force ε^* to be very small and render the regime $\varepsilon < \varepsilon^*$ essentially inaccessible. A natural and low-cost improvement is to redefine ε^* as half the minimum distance between *distinct* observations, and to argue that the cardinality of accepted permutations is then constant (rather than at most one). Theorem 1's conclusion would then extend to that setting with a minor reweighting. This was explicitly suggested during a previous round of reviews.

M8. Narrow scope of LFI benchmark.

Beyond ABC and ABC-Gibbs, the likelihood-free inference landscape now includes neural posterior/likelihood estimation (e.g. SNPE/SNLE), simulation-based inference via normalizing flows, and generalised Bayesian methods that operate on discrepancies closely related to the one used here. The current comparison—against Vanilla ABC, ABC-PMC, ABC-Gibbs—does not position permABC within this wider landscape. I do not require a full battery of NN-based methods, but a short discussion, and ideally one numerical point on the Gaussian toy model, would help the reader gauge when permABC is the right tool.

Minor concerns

m1. Section 2 starts with a `\subsection` and no introductory text. The reader jumps from the section heading directly into "State of the art." A brief paragraph outlining the three strategies would help.

m2. The Gaussian toy model (Section 3.2) never specifies the values of (a, b, s, n, K). The reader is told this is "a classical hierarchical model" but the exact experimental setup (prior hyperparameters, sample size per compartment, values of K tested) is not given in the main text.

m3. Figure 2 (Over-Sampling posterior evolution) is never properly discussed. The caption says "Gaussian toy model" without specifying which parameters. More importantly, the text around Figure 2 now contains the formal $M \to \infty$ analysis but the figure is not interpreted in light of that analysis. The prior-like behavior of β for large M and the concentration of the local parameters should be explicitly pointed out on the figure.

m4. The ABC-Gibbs comparison (Section 3.3) uses a deliberately adversarial model. The ridge structure in the over-parameterized model is designed to make Gibbs fail. While the point is valid, a fairer comparison would also show a model where ABC-Gibbs works well (e.g. the standard Gaussian toy model of Section 3.2 with weak dependencies), to demonstrate that the methods have complementary strengths rather than presenting a one-sided comparison.

m5. Tracked-change markup (`\old{}`, `\new{}`) is still visible in the compiled PDF (rendered as red strikethrough and blue text). These must be resolved before submission.

m6. Notation overload on M. In Section 2.1, M denotes the number of datasets simulated per parameter value. In Section 2.3 (Over-Sampling), M denotes the number of simulated compartments. The reuse is potentially confusing and should be disambiguated.

m7. The "unique particle rate" (Section 2.1) is proposed as a complementary diagnostic for sample diversity but is never reported in any experiment. Either use it or remove the discussion.

m8. Missing discussion of computational cost. The $\mathcal{O}(K^3)$ cost of the assignment step is mentioned but never benchmarked. For $K = 94$ (SIR), what fraction of the total runtime is spent in the assignment step versus simulation? This matters for practitioners deciding whether to adopt the method. The revised Section 2.3 on warm-started swaps claims an $\mathcal{O}(K^2)$ amortised cost; this should be substantiated by a quantitative comparison between full LSA and the proposed warm-swap scheme, ideally reporting both runtime and the fraction of iterations for which the returned permutation coincides with the exact optimum.

m9. Hyperparameters of the Gaussian toy model are not reported. Section 3.2 refers to "a classical hierarchical model" without specifying (a, b, s, n) or the values of K tested. These should appear in the main text or in the figure captions.

m10. Initialization of permABC-OS. Appendix C.2 calibrates ε as a function of M_0 under a fixed simulation budget. It is unclear what happens when the budget forces M_0 to be too small for the target ε^*. Would importance-based proposals for θ (rather than prior sampling) help? This has practical implications.

m11. Role of the critical tolerance ε^* in applications. The small-ε regime where permABC and standard ABC coincide relies on $\varepsilon < \varepsilon^*$, but for realistic data ε^* can be extremely small (a few nearly-duplicate compartments suffice). The paper should discuss how the practitioner is expected to behave when $\varepsilon > \varepsilon^*$: which of the two pseudo-posteriors is preferable, and on what grounds?

m12. Complex epidemiological models. The SIR application is simple relative to what is common in modern epidemiology (SEIHR, age structure, delayed observation). A discussion of scalability beyond the current three local parameters per compartment would strengthen the case for real-world adoption.

Questions for the authors

1. Can you provide posterior quality metrics (KL, W_2, coverage) for the Gaussian toy model where the posterior is available in closed form?
2. What happens to permABC when exchangeability is only approximate (e.g. mildly heterogeneous priors across compartments)?
3. For the SIR application, have you run a simulation study with known ground truth to validate the method before applying it to real data?
4. The asymptotic material connecting permABC to WABC (compartment-level Wasserstein viewpoint, $K \to \infty$ limit) seems highly relevant. Why was it not included?
5. Can you provide a closed-form normal–normal example $\mu_k \sim N(\beta, 1)$, $y_k \sim N(\mu_k, 1)$ in which the behaviour of permABC-OS as a function of M can be compared analytically with standard rejection ABC (for which Bornn et al. (2017) show that increasing M still yields samples from π_ε)? This would clarify whether the effect of M is specific to permABC.
6. Do the tracked-change macros (`\new`, `\old`) correspond to all substantive changes made since the previous version? A summary of modifications would help subsequent rounds of review.

Assessment

The core idea of permABC is sound, practically motivated, and computationally elegant. The Over-Sampling and Under-Matching extensions are original contributions. However, the paper needs:

1. at least one proper posterior quality metric beyond ε,
2. a comparison with standard ABC-SMC (without permutations),
3. either formal statements of the $M \to \infty$ results or integration of the existing asymptotic material,
4. a simulation-based validation of the SIR model (synthetic data with known truth),
5. a numerical benchmark against WABC (Hilbert/Sinkhorn approximations) on the Gaussian toy model,
6. an extension of Lemma 1 / Theorem 1 to the case of ties,
7. a quantitative substantiation of the $\mathcal{O}(K^2)$ warm-swap claim of Section 2.3.

With these additions, the paper would make a solid contribution to JASA.